# 计算系统生物学作业4

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**1 Waiting times for chemical reactions**

a.

证明. We need to check

$$\int_0^\infty p(\tau)d\tau = 1 \tag{1}$$

We have $\int_0^\tau p(\tau)d\tau = \int_0^\infty re^{-r\tau}d\tau = -\int_0^\tau e^{-r\tau}d(-r\tau) = -e^{-r\tau}\Big|_0^\infty = 1$ □

b.

证明. From

$$\theta(u) = \frac{1}{r}\ln(\frac{1}{u}) \tag{2}$$

We get

$$u = e^{-r\theta} \tag{3}$$

□

According to the formula

$$p(\theta) = p(u)\left|\frac{du}{d\theta}\right| \tag{4}$$

We have

$$p(\theta) = 1|(-r)e^{-r\theta}| = re^{-r\theta} \tag{5}$$

**2 Stochastic simulation of an auto-regulatory system**

a. For the ODE

$$\frac{dx}{dt} = \beta - \alpha x \tag{6}$$

i.e.

$$\frac{dx}{dt} + \alpha x = \beta \tag{7}$$

This is a first-order linear differential equation. The integrating factor is $e^{\alpha t}$. Here we assume $\alpha \neq 0$, we get

$$x(t) = \frac{\beta}{\alpha} + (x_0 - \frac{\beta}{\alpha})e^{-\alpha t} \tag{8}$$

Take $\beta = 6, \alpha = 3$, we have

$$x(t) = 2 + (x_0 - 2)e^{-3t} \tag{9}$$

b. 1. $P(x = 2|t = 1) = \frac{\alpha x}{\beta+\alpha x} = \frac{3\alpha}{\beta+3\alpha}$

2. It successfully describes the average behavior during the stochastic process, or the expectation value, instead of the behavior of one single experiment.

c. The codes are as in 1:

d. 1. For system A

$$\frac{dx}{dt} = A - \gamma x \tag{10}$$

Set $\frac{dx}{dt} = 0$, we get fixed point $x^* = \frac{A}{\gamma}$, and it is stable, since $f'(x) = -\gamma$.

For system B

$$\frac{dx}{dt} = B\frac{K}{K+x} - \gamma x \tag{11}$$

**Algorithm 1** Gillespie First-Reaction Simulation

**procedure** GillespieFirstReaction($x_0$, $\alpha$, $\beta$, $T_{\max}$)
    $t \leftarrow 0$
    $x \leftarrow x_0$
    $R \leftarrow [(t, x)]$
    **while** $t < T_{\max}$ **do**
        $k_{\text{trans}} \leftarrow \beta$
        $k_{\text{deg}} \leftarrow \alpha x$
        Draw $u_1, u_2 \sim \text{Uniform}(0, 1)$
        $\tau_{\text{trans}} \leftarrow -\ln(u_1)/k_{\text{trans}}$
        **if** $k_{\text{deg}} > 0$ **then**
            $\tau_{\text{deg}} \leftarrow -\ln(u_2)/k_{\text{deg}}$
        **else**
            $\tau_{\text{deg}} \leftarrow \infty$
        **end if**
        **if** $\tau_{\text{trans}} < \tau_{\text{deg}}$ **then**
            $t \leftarrow t + \tau_{\text{trans}}$
            $x \leftarrow x + 1$
        **else**
            $t \leftarrow t + \tau_{\text{deg}}$
            $x \leftarrow x - 1$
        **end if**
        $R$.append($(t, x)$)
    **end while**
    **return** $R$
**end procedure**

Set $\frac{dx}{dt} = 0$, we get fixed point

$$x^* = \frac{-K + \sqrt{K^2 + \frac{4BK}{\gamma}}}{2} \tag{12}$$

and it is stable since $f'(x) = -\frac{BK}{(K+x)^2} - \gamma$

2. $k_{birth} = A, k_{death} = \gamma x$,hence

$$x(t) = \frac{A}{\gamma} + (x_0 - \frac{A}{\gamma})e^{-\gamma t} \tag{13}$$

Take $\gamma = 1$

$$x(t) = A + (x_0 - A)e^{-t} \tag{14}$$

At steady state

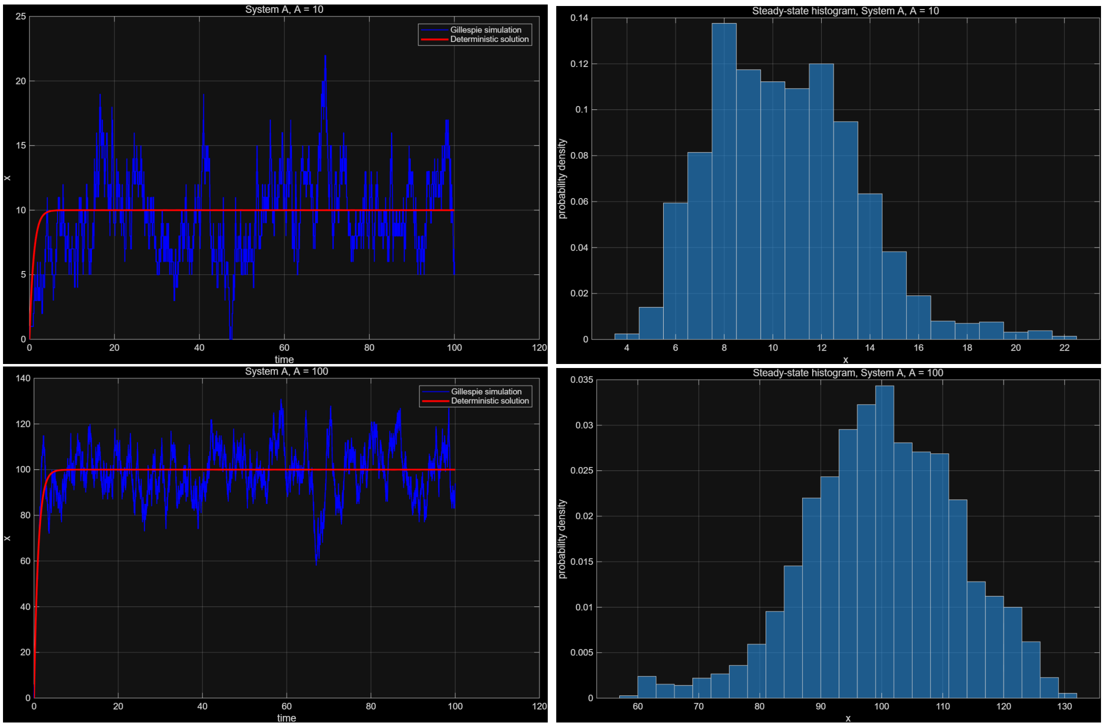
$$< x >= \frac{A}{\gamma} = A \tag{15}$$

Notice that at steady state, we have Poisson distribution

$$x \sim \text{Poisson}(\frac{A}{\gamma}) \tag{16}$$

hence mean=$\frac{A}{\gamma}$, variance=$\frac{A}{\gamma}$

Take A=10/100,$\gamma = 1$




Listing 1: Output of System A simulation

```
System A, A = 10
Mean = 10.504
Variance = 9.066
Poisson prediction: mean = variance = 10.000
```

```
System A, A = 100
Mean = 99.638
Variance = 155.505
Poisson prediction: mean = variance = 100.000
```

3. For system B, $k_{birth} = B\frac{K}{K+x}, k_{death} = \gamma x$
Notice that

$$B\frac{K}{K+x^*} = \gamma x^* \tag{17}$$

Take $x^* = 100, K = 100, \gamma = 1$, we have $B = 200$.
For the same mean value, we compare A with B on variance. Denote

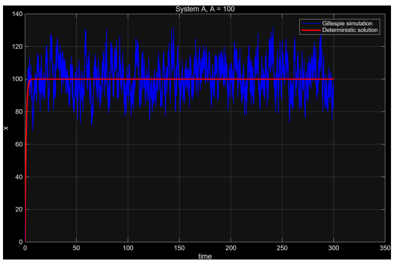
$$f(x) = B\frac{K}{K+x} - \gamma x$$

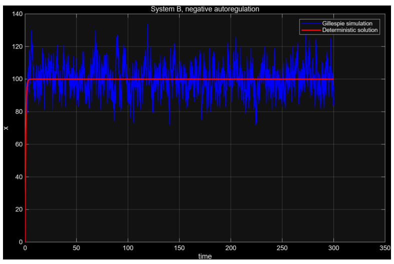
$$f'(x) = -\frac{BK}{(K+x)^2} - \gamma \tag{18}$$

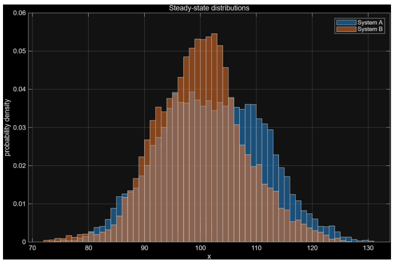
Hence $f'(100) = -1.5$. For linear noise, we have

$$\text{Var}(x) = \frac{B\frac{K}{K+x^*} + \gamma x^*}{-2f'(x^*)} = \frac{200\frac{100}{200} + 100}{3} = 66.67 \tag{19}$$

We have $Var_B(x) < Var_A(x)$.




Listing 2: Comparison between A and B

```
Comparison at same mean
System A: mean = 101.643, variance = 86.715
System B: mean = 99.577, variance = 63.984
Expected: System A variance around 100
Expected: System B variance around 66.7
```

## 3 Deterministic and stochastic description of a system with nonlinear degradation

a. Since X is indistinguishable, one enzymatic degradation can occur in $\binom{n}{2} = \frac{n(n-1)}{2}$ ways.

b.

$$\frac{dP_n}{dt} = kP_{n-1} + \gamma\binom{n+1}{2}P_{n+1} - [k + \gamma\binom{n}{2}]P_n \tag{20}$$

i.e.

$$\frac{dP_n}{dt} = kP_{n-1} + \gamma\frac{(n+1)n}{2}P_{n+1} - [k + \gamma\frac{n(n-1)}{2}]P_n \tag{21}$$

where we set $P_{-1} = 0$ for convenience.

c.

$$< n >= \sum_{n=0}^{\infty} nP_n \tag{22}$$

Hence we have

$$\frac{d < n >}{dt} = k - < \frac{\gamma n(n-1)}{2} > \tag{23}$$

i.e.

$$\frac{d < n >}{dt} = k - \frac{\gamma}{2}(< n^2 > - < n >) \tag{24}$$

Notice that $V =< n^2 > - < n >^2$, we have

$$\frac{d < n >}{dt} = k - \frac{\gamma}{2}(V + < n >^2 - < n >) \tag{25}$$

Hence

$$\frac{d < n >}{dt} = k - \frac{\gamma}{2}(< n >^2 - < n >) + \frac{\gamma}{2}V \tag{26}$$

d. We can simplify by considering that $< n^2 > \approx < n >^2$, i.e. $V \approx 0$. Hence we have

$$\frac{dx}{dt} = k - \frac{\gamma}{2}x(x-1) \approx k - \frac{\gamma}{2}x^2 \tag{27}$$

Hence

$$x^* = \sqrt{\frac{2k}{\gamma}} \tag{28}$$

e. Notice that

$$0 = k - \frac{\gamma}{2}(< n >^2 - < n >) - \frac{\gamma}{2}V \tag{29}$$

Hence

$$< n >^2 - < n > + V = \frac{2k}{\gamma} \tag{30}$$

We require $V \ll < n >^2 - < n >$, i.e. $V \ll \frac{2k}{\gamma}$, i.e. $\frac{\sqrt{V}}{<n>} \ll 1$.